Networking for C++29: A Call to Action

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Networking for C++29: A Call to Action

Abstract

Twenty-one years is long enough.

The Network Endeavor (P4100R0[1]) defines the work: eleven papers, two stages, two libraries shipping today. This paper proposes how to do the work - a production pipeline, a continuous workflow, and an open invitation to the people whose expertise can make it succeed. The committee has the talent. The implementations exist. The only missing ingredient is coordination.

Revision History

R0: March 2026 (post-Croydon mailing)

- Initial version.

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1. The Moment

The committee has been trying to standardize networking since N1925[2] (2005). The Networking TS reached publication in 2018. It was not merged. The executor unification effort consumed a decade. Networking waited.

The committee polled twice on whether a single async model should govern the entire standard library. Neither poll achieved consensus (P2453R0[3]). The committee has two async models - `std::execution` ships in C++26, and `std::execution::task` is both a coroutine and a sender. The price of two models has already been paid.

C++29 is the target. The Network Endeavor (P4100R0[1]) defines the work: eleven papers backed by two shipping libraries - Capy and Corosio - with three independent adopters. The companion papers document the technical foundations: coroutine properties (P4088R0[4]), domain separation (P4099R0[5]), and sender-coroutine bridges (P4092R0[6], P4093R0[7]).

The work is defined. What remains is the question every large institution eventually faces: can we organize ourselves to deliver?

This paper is the Network Endeavor's organizational proposal. It is authored by the endeavor's architects. The pipeline, the teams, and the workflow are our design. The invitation to the `std::execution` architects and to the broader committee is genuine - the endeavor cannot succeed without their expertise, and their requirements must be addressed. But this is our proposal, not a jointly authored plan. We are asking the committee to let us do the work, with every stakeholder's requirements visible, documented, and addressed at every stage.

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2. Prerequisites

Two agreements must hold before the pipeline described in this paper can function. Both are the subject of companion papers. This paper does not relitigate them - it assumes they are accepted and builds on them.

2.1 Domain Partition

Coroutine-native I/O serves byte-oriented networking. `std::execution` serves GPU dispatch, heterogeneous execution, and compile-time work graphs. Two models, each correct for its domain. The technical basis is documented in P4088R0[4] and P4099R0[5].

Domain specialization is not fragmentation. C++ has multiple container types, multiple string types, multiple smart pointer types. GPU compute has `nvexec` with CUDA extensions and a separate namespace. Two async models for two distinct domains is the same principle the committee has applied throughout the standard.

2.2 Interoperability Commitment

Both models must interoperate. Coroutine-native code must be able to consume senders. Sender-based code must be able to consume awaitables. The bridge papers (P4092R0[6], P4093R0[7]) have explored the mechanisms. The cost is acknowledged by both sides: the coroutine side accepts that interop is a hard requirement; the sender side accepts that coroutine-native I/O makes different trade-offs - frame allocation, type erasure, ABI stability - that serve networking's specific needs.

2.3 Architectural Coherence

The Networking TS was compromised by successive design demands that eroded its architectural coherence (P4094R0[8]). The stream model, the buffer sequences, and the executor architecture were each sound. The accumulated weight of unification requirements - each individually reasonable - produced a design that could not ship.

This endeavor maintains structural safeguards to prevent that pattern from recurring. Section 4 describes them.

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3. A New Way of Working

The committee's traditional workflow is meeting-centric: papers are published in mailings, discussed at plenary meetings three times a year, and revised between meetings by their authors. For a single paper, this works. For an eleven-paper series with cross-cutting dependencies, it is too slow.

This paper proposes a continuous workflow that operates between meetings and delivers visible progress at every stage.

3.1 Communication

- **Mattermost** - A dedicated instance with channels for each team (Section 4). Real-time collaboration, asynchronous by default. No mandatory meetings. Synchronous calls when a team needs them, not on a fixed schedule.
- **SG4 and LEWG mailing lists** - Formal discussion of design decisions that affect the committee's direction.
- **The Reflector** - Committee-wide announcements and cross-team coordination.

3.2 Artifacts

- **A shared GitHub repository** for papers, design documents, specs, and rationale. Every artifact is version-controlled and publicly visible.

- **Non-WG21 contributors** can participate in testing, wording review, and implementation. The repository is open.
- **Every design decision** is documented with its rationale - including dissenting views and the reasons disagreements were resolved. The record is complete.

3.3 Implementation

- **The Beman Project** serves as the reference implementation vehicle. As papers move through the pipeline, the implementation team contributes to Beman continuously.
- **Cross-platform CI**, test suites, and benchmarks are available to the committee at every stage. The committee can clone the repo and run the tests.

3.4 Knowledge Preservation

- **WG21 SAGE** captures the reasoning behind every decision, so future revisits have full context.
- Rationale is documented at every decision point. Not just what was decided, but why - and what alternatives were considered and rejected.
- Institutional knowledge is a first-class deliverable alongside the feature itself. When this endeavor is complete, the committee will have not just networking in the standard, but a complete record of how it got there.

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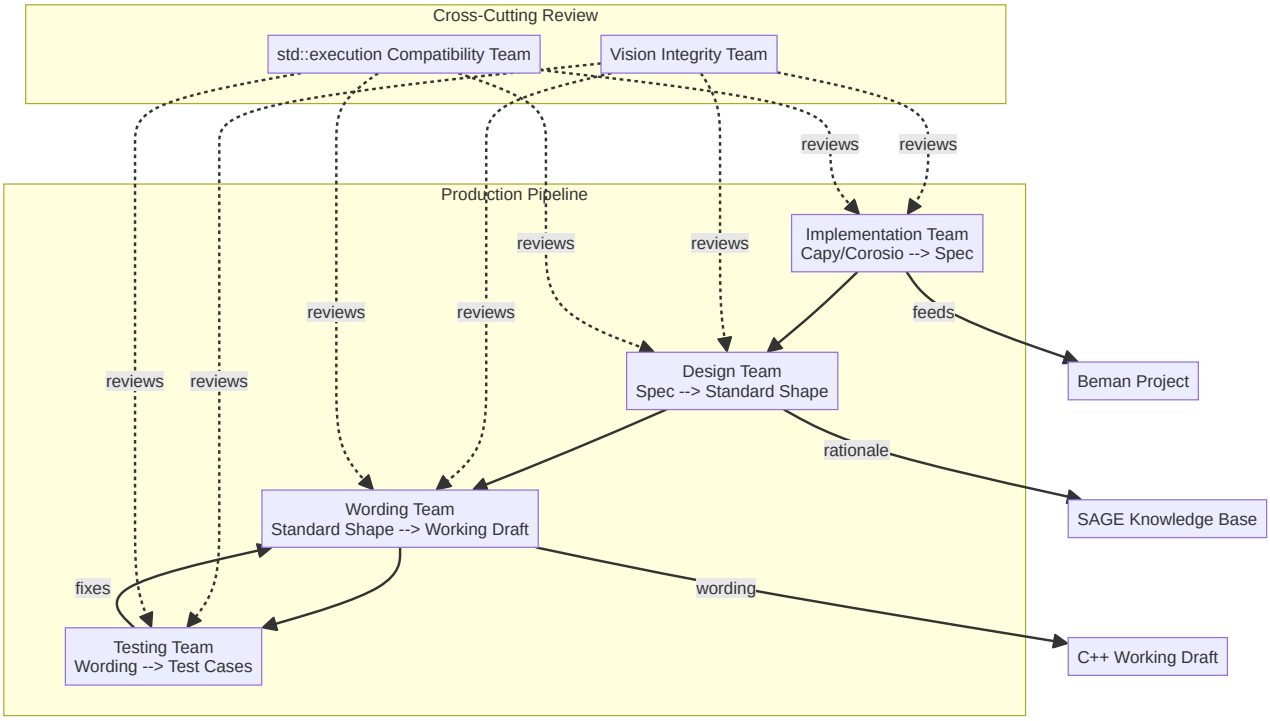
4. The Pipeline

Six teams. Four in a sequential production pipeline. Two cross-cutting review teams.



4.1 Implementation Team

Examines Capy and Corosio - the working code. Writes a spec: what the libraries do, how they do it, what the interfaces are, what the invariants are. The spec is a faithful description of deployed behavior, not a design document. The spec is the input to the Design Team.

4.2 Design Team

Takes the spec and determines the standard shape. What belongs in `std`? What are the concepts, the type requirements, the customization points? Where does the design differ from the implementation and why? Sender interop, allocator policy, error model, and ABI boundaries are decided here.

4.3 Wording Team

Takes the standard shape and produces formal wording for the C++ Working Draft. Clause structure, stable names, cross-references, editorial conventions. This is LWG-quality prose.

4.4 Testing Team

Takes the wording and debugs it. Writes test cases against the wording - not the implementation. Finds ambiguities, contradictions, and missing edge cases. This team is open to non-WG21 members. Anyone who can read a specification and write a test is welcome.

4.5 Vision Integrity Team

Ensures the result preserves the architectural properties that make the design work: type erasure, ABI stability, separate compilation, and the three-layer architecture documented in P4100R0[1] Section 6. Reviews every output of the pipeline for architectural coherence.

The Networking TS lost its architectural coherence through successive compromises, each individually reasonable (P4094R0[8]). This team exists so that does not happen again. The Capy and Corosio architects staff this team.

4.6 `std::execution` Compatibility Team

Ensures every artifact produced by the pipeline has a clean bridge to `std::execution`. Reviews the spec, the standard shape, and the wording at every stage. Guarantees that coroutine-native awaitables can be consumed from sender-based code and vice versa. Documents the bridge cost at each boundary.

The reciprocal commitment: the pipeline teams accept that interoperability with `std::execution` is a hard requirement and design for it. The Compatibility Team accepts that coroutine-native I/O makes different trade-offs and does not require that networking adopt sender semantics internally. Both sides acknowledge the cost. Both sides get a voice.

The P2300 architects staff this team.

4.7 Continuous Production

Each team works paper by paper. The eleven papers flow through the pipeline. Multiple papers occupy different stages simultaneously - Paper 1 in Wording while Paper 2 is in Design while Paper 3 is in Implementation. Every handoff produces a visible artifact in the GitHub repository.

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5. The People

WG21 has enormous talent. The following is our best assessment of the expertise available in the committee and the broader community. No one is assigned. Everyone is invited. These tables reflect the author's understanding of where each person's strengths align with the work. They are a starting point, not a roster. Anyone who wants to contribute to any team is welcome - the only criterion is willingness.

5.1 Vision Integrity Team

Person	Affiliation	Demonstrated Strengths
Vinnie Falco	C++ Alliance	Capy/Corosio architect; Network Endeavor author
Steve Gerbino	C++ Alliance	Corosio co-author; constructed comparisons
Mohammad Nejati	C++ Alliance	Corosio co-author; platform backends
Mungo Gill	C++ Alliance	Capy co-author; frame allocator propagation
Michael Vandeberg	C++ Alliance	Corosio co-author

5.2 `std::execution` Compatibility Team

Person	Affiliation	Demonstrated Strengths
Eric Niebler	NVIDIA	Ranges, sender composition, structured concurrency
Michał Dominiak	NVIDIA	P2300 lead author; type erasure and language feature expertise
Lewis Baker	(ind.)	Coroutines, symmetric transfer; cppcoro, folly::coro
Lee Howes	(ind.)	Async infrastructure, futures, system execution context
Kirk Shoop	Microsoft	Sender algorithms and reactive composition patterns
Michael Garland	NVIDIA	GPU parallel programming; heterogeneous dispatch
Bryce Adelstein Lelbach	NVIDIA	Parallel algorithms, mdspan; LEWG process experience
Mark Hoemmen	NVIDIA	P2300 steward; linalg, mdspan; cross-domain integration
Dietmar Kühl	Bloomberg	Sender networking (P2762), task design (P3552)

5.3 Implementation Team

Person	Affiliation	Demonstrated Strengths
Chris Kohlhoff	(ind.)	Asio architect; twenty years of stream model and buffer design
Steve Gerbino	C++ Alliance	Corosio co-author; constructed comparisons across designs

Person	Affiliation	Demonstrated Strengths
Mohammad Nejati	C++ Alliance	Corosio co-author; platform backends (IOCP, epoll, kqueue)
Mungo Gill	C++ Alliance	Capy co-author; frame allocator propagation
Michael Vandeberg	C++ Alliance	Corosio co-author
Klemens Morgenstern	(ind.)	Boost.Cobalt author; cross-library bridge experience
Ruben Perez	(ind.)	Boost.MySQL; independent adopter migrating to Corosio
Lee Howes	(ind.)	folly::coro and system execution context; production scale
Lewis Baker	(ind.)	cppcoro, folly::coro; symmetric transfer and coroutine internals

5.4 Design Team

Person	Affiliation	Demonstrated Strengths
Eric Niebler	NVIDIA	Ranges, sender composition; first-principles API design
Michał Dominiak	NVIDIA	P2300 lead author; type erasure and language feature expertise
Kirk Shoop	Microsoft	Sender algorithms and reactive composition patterns
Dietmar Kühl	Bloomberg	Sender networking (P2762), task design (P3552); bridge experience
Peter Dimov	(ind.)	Minimal abstractions; Boost library design; allocator expertise
John Lakos	Bloomberg	Physical design, allocators, levelization; dependency architecture
Howard Hinnant	(retired)	Move semantics, chrono, libc++; former LWG Chair
Jonathan Müller	think-cell	Coroutine safety analysis, task design
Michael Garland	NVIDIA	GPU parallel programming; heterogeneous dispatch requirements
Mark Hoemmen	NVIDIA	P2300 steward; linalg, mdspan; cross-domain integration
Bryce Adelstein Lelbach	NVIDIA	Parallel algorithms, mdspan; LEWG process experience
Ville Voutilainen	Qt Company	EWG Chair; language evolution; allocator review

Person	Affiliation	Demonstrated Strengths
Andrzej Krzemieński	(ind.)	Contracts, error handling, postconditions; error model design

5.5 Wording Team

Person	Affiliation	Demonstrated Strengths
Nina Ranns	(ind.)	CWG expertise; WG21 Secretary; contracts and allocator wording
Dietmar Kühl	Bloomberg	Extensive wording experience (P2762, P3552)
Nicolai Josuttis	(ind.)	Standard library wording; teachability and clarity
Michał Dominiak	NVIDIA	P2300 wording author; editorial precision
Inbal Levi	Microsoft	LEWG Chair; process coordination
David Sankel	Adobe	Beman Project lead; library wording experience

5.6 Testing Team

Person	Affiliation	Demonstrated Strengths
Steve Gerbino	C++ Alliance	Constructed comparisons; systematic testing
Ruben Perez	(ind.)	Independent adopter; real-world migration testing
Klemens Morgenstern	(ind.)	Cross-library interop testing
Nicolai Josuttis	(ind.)	Teachability testing; can the wording be understood?
(open)		Non-WG21 contributors welcome: anyone who can read a spec

People appear on multiple teams where their expertise spans boundaries. That is intentional. The teams are work areas, not silos.

6. Corporate Stakeholders

Networking in C++ affects every company that deploys C++ at scale. This endeavor explicitly invites corporate stakeholders to put their requirements on the table - openly, at every stage.

Organization	Stake
NVIDIA	Largest `std::execution` investment; GPU dispatch, heterogeneous execution
Meta	folly::coro deployed at billion-user scale; production async infrastructure
Google	`Co<T>` coroutine type; production safety constraints
Bloomberg	Physical design, allocators, P3552 task; ABI stability and separate compilation
Microsoft	Sender algorithms, reactive extensions, compiler implementation
Apple	Standard library implementation (libc++); implementation perspective
C++ Alliance	Capy, Corosio, Boost libraries; implementation and deployment experience

If a company needs something from the design, they say so. The requirement is documented, visible, and addressed. No hidden agendas. Every stakeholder's cards are on the table.

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7. The Pipeline in Motion

P4100R0[1] Section 11 defines the timeline: all eleven papers through LEWG by end of 2028, LWG wording through 2029H1. The pipeline maps onto that timeline.

Multiple papers occupy different pipeline stages simultaneously. While Paper 1 (Coroutines for I/O) is in Wording review, Paper 2 (I/O Buffer Ranges) can be in Design, and Paper 3 (Stream Concepts) can be in Implementation. The Beman Project receives implementation output continuously. Each handoff produces a visible artifact in the GitHub repository.

The eleven papers divide into two stages that mirror the library split:

Stage One (Papers 1-4) - Pure C++20 abstractions. No platform dependency. Coroutine execution protocol, buffer concepts, stream concepts, coroutine runtime. These enable sans-I/O protocols in the ecosystem.

Stage Two (Papers 5-11) - Platform I/O. Timers, signals, files, TCP, DNS, UDP, TLS. The first three papers in Stage Two have nothing to do with networking. The committee can standardize seven papers before anyone says the word "socket."

If C++29 proves too aggressive, the Stage One papers deliver standalone value for C++32. The committee ships what is ready. Each paper that completes the pipeline is a shippable unit - the endeavor does not require all eleven papers to succeed for any of them to succeed. Partial delivery is still delivery. The ecosystem benefits from coroutine execution protocol and buffer concepts whether or not TLS makes the same standard.

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8. The Stakes

This is not just about shipping a feature.

The committee has been perceived - by its own members, by the broader C++ community, and by observers in other language ecosystems - as unable to coordinate large efforts. Networking is the longest-running example. Twenty-one years from first proposal to the present, and the standard has no sockets, no DNS, no TLS.

The talent exists. The implementations exist. The technical foundations are documented. The bridge mechanisms are explored. What has been missing is a way of working that lets the committee's collective expertise converge on a shared goal instead of diverging into competing visions.

If this endeavor succeeds, the committee will have delivered networking for C++29. That alone justifies the effort. But the committee will also have demonstrated something larger: that WG21 can organize a multi-year, multi-team, cross-faction effort and ship the result. That is a model the committee can use again - for reflection, for pattern matching, for whatever comes next.

The work is defined. The people are here. The question is whether we choose to do this together.

Acknowledgments

The authors of `std::execution` - Eric Niebler, Kirk Shoop, Lewis Baker, Lee Howes, Michał Dominiak, Michael Garland, Bryce Adelstein Lelbach, and their collaborators - invested years in a principled async framework for C++. Their work on sender/receiver serves domains where compile-time composition and heterogeneous dispatch matter. This endeavor builds alongside that work and depends on their expertise for interoperability.

Chris Kohlhoff designed Asio's stream model, buffer sequences, and executor architecture. Twenty years of production deployment is the foundation this work builds on.

The committee designed C++20 coroutines. Gor Nishanov, Lewis Baker, and their collaborators gave C++ the language mechanisms that make coroutine-native I/O possible.

Dietmar Kühl's work on sender-based networking (P2762[9]) and the coroutine task type (P3552[10]) explored the design space from the sender side. That exploration informs the bridge design.

Mark Hoemmen's stewardship of `std::execution` through C++26 demonstrated the sustained effort required to shepherd a large proposal through the committee process.

The Beman Project, led by David Sankel, provides the reference implementation infrastructure that makes continuous production possible.

Peter Dimov, Andrzej Krzemieński, John Lakos, Howard Hinnant, Nina Ranns, Nicolai Josuttis, Jonathan Müller, Ville Voutilainen, and Inbal Levi bring decades of library design, wording, and process expertise that this endeavor cannot succeed without.

Ruben Perez, Marcelo Zimbres Silva, Klemens Morgenstern, and the Boost.Postgres team adopted Capy and Corosio independently. Their experience is evidence that cannot be manufactured.

Steve Gerbino, Mohammad Nejati, Mungo Gill, and Michael Vandeberg co-authored the libraries that make this proposal concrete rather than aspirational.

References

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